

TARUN SAILESH VANGIPURAPU

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PROFESSIONAL SUMMARY

Software Development Engineer with 5+ years of experience building and enhancing test automation frameworks, distributed systems, and scalable backend services across enterprise environments. Expert programming skills in Java and Python with hands-on experience across the full software development lifecycle, including CI/CD pipeline development, distributed version control with Git, and RESTful web services integration. Proven ability to develop automation solutions, support continuous improvement efforts, and architect high-availability systems with strong data integrity and scalability principles. Comfortable thriving in fast-paced, cross-functional organizations, debating ideas openly, and pivoting on continuously evolving requirements. Strong written and verbal communication skills with the ability to clearly explain technical decisions and write technical specification documents.

EXPERIENCE

Quadrant Technologies | Software Engineer

Jan 2024 – Present

- Developed and enhanced automation frameworks and backend microservices using Java and Spring Boot to process high-volume telemetry and operational data streams, enabling real-time analytics and system monitoring.
- Designed and implemented event-driven processing pipelines using Azure Event Hubs and publish/subscribe messaging services, enabling scalable ingestion and distributed processing of large volumes of platform activity data.
- Built and maintained CI/CD pipelines using Git-based workflows and automated build processes, supporting continuous improvement efforts and reducing manual deployment steps across multiple service deployments.
- Architected RESTful APIs and web services integrations using Spring Boot and Spring WebFlux, supporting real-time communication between distributed microservices and ensuring high availability and easy maintainability.
- Developed React.js user interfaces with Redux and modern component architecture, enabling interactive dashboards and improved user experience for operational monitoring tools.
- Implemented OAuth2 and JWT-based authentication using Spring Security, strengthening API security and enabling secure service-to-service communication across distributed systems.
- Integrated caching strategies using Redis and optimized database queries to improve application performance, scalability, and reduce latency for frequently accessed endpoints.
- Designed and deployed containerized microservices using Docker and Kubernetes (AKS), enabling scalable and reliable cloud-native deployments with consistent Unix/Linux-based infrastructure.
- Implemented observability and monitoring using Azure Monitor, Prometheus, and logging frameworks, improving system visibility and enabling faster troubleshooting and debugging of distributed services.
- Designed and developed ML evaluation solutions to assess model output quality, enabling data-driven insights into system performance and supporting iterative improvement of machine learning pipelines.
- Collaborated closely with cross-functional teams including data scientists, product managers, and QA engineers to define test strategies, review automation requirements, and align deliverables with evolving business goals.
- Authored technical specification documents and design proposals for new automation modules, facilitating open team discussions and enabling informed architectural decisions across distributed engineering teams.
- Applied Unix/Linux administration skills to configure and manage server environments, automate shell-based workflows, and support infrastructure operations for deployed microservices.
- Conducted code reviews and mentored junior engineers on best practices in automation framework development, clean code principles, and distributed systems design patterns.

Wipro Pvt Ltd | Software Engineer

March 2021 – Jan 2023

- Developed full-stack enterprise applications using Java, Spring Boot, and React.js, enabling scalable web platforms that supported internal business workflows and operational dashboards across the full software development lifecycle.
- Built and enhanced RESTful APIs and web services integrations using Spring Boot and Spring MVC, supporting distributed systems integration and cross-service communication.
- Developed microservices-based backend components using Spring Boot and Spring MVC, improving modularity, maintainability, and scalability of the application architecture with a focus on data integrity.
- Implemented event-driven processing using Apache Kafka and publish/subscribe messaging services, enabling reliable asynchronous communication between distributed services.
- Applied Spring Security with OAuth2 and JWT authentication to secure APIs and enforce role-based access control, ensuring high availability and security across enterprise applications.
- Optimized application performance through Redis caching strategies and database query tuning in SQL Server, improving API response times and reducing database load at scale.

- Implemented and supported CI/CD pipelines using Jenkins and Git-based workflows, automating build validation, testing, and continuous deployment processes for microservices across multiple environments.
- Containerized applications using Docker and deployed services on Kubernetes, enabling scalable and consistent cloud-native application deployments on Linux-based infrastructure.
- Applied JUnit, Mockito, and SonarQube for unit testing, automation framework support, code quality, and static analysis, contributing to continuous improvement of software quality standards.
- Developed and maintained reusable test automation framework components using Java and Python, reducing test authoring time and increasing overall test coverage across core application modules.
- Designed and executed integration and regression test suites for distributed service endpoints, identifying and resolving defects early in the development lifecycle and improving release confidence.
- Leveraged Git-based distributed version control workflows including branching strategies, pull request reviews, and merge conflict resolution to maintain code quality and team collaboration at scale.
- Participated in Agile sprint ceremonies including planning, retrospectives, and daily standups, contributing to transparent communication and consistent delivery of features and automation improvements.
- Administered and configured Unix/Linux server environments to support local development, automated testing pipelines, and staging deployments, ensuring environment consistency across teams.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, SQL, Bash/Shell

Automation & Testing: Test Automation Frameworks, JUnit, Mockito, SonarQube, ML Evaluation Solutions

CI/CD & DevOps: CI/CD Pipelines, Jenkins, Git, GitHub, GitLab, Distributed Version Control, Containerization

Backend Development: Spring Boot, Spring MVC, REST APIs, Web Services, Microservices, Distributed Systems

Cloud & Infrastructure: Microsoft Azure, AWS, Docker, Kubernetes, Linux/Unix Administration

Databases: PostgreSQL, MySQL, MongoDB, NoSQL, SQL Server, Redis

Frontend: React.js, Redux, HTML, CSS

API & Integration: RESTful APIs, Publish/Subscribe Messaging, Apache Kafka, Azure Event Hubs, Swagger/OpenAPI

EDUCATION

University of North Texas Denton, TX

Master of Science in Information Systems

Gitam University Hyd, TS

Bachelor of Technology in Electronics and Communication Engineering